Anagha R S

anaghars11165@gmail.com | +91 97780 61676 | Kollam, Kerala | LinkedIn | GitHub

Objective

Computer Science postgraduate with experience in software engineering, backend development, and AI/ML applications. Skilled in Java, Python, SQL, Object-Oriented Programming (OOP), FastAPI, Git/GitHub, and AWS. Passionate about building scalable software solutions, solving complex problems, and continuously learning new technologies in collaborative development environments.

Education

Cochin University of Science and Technology (CUSAT) — Kochi, Kerala
M.Tech — CS & Engineering, Data Science & AI | GPA: 8.8/10 — *Aug 2024 – Apr 2026*

Government Engineering College, Idukki — Idukki, Kerala
B.Tech — Computer Science & Engineering | GPA: 8.04/10 — *Nov 2020 – Jun 2024*

Experience

Indian Institute of Technology Madras — Chennai, Tamil Nadu
AI/ML Research Intern — *Dec 2025 – May 2026*

- Developed a custom **ResNet1D** model in PyTorch, achieving **99%+ accuracy** on the CWRU benchmark.
- Compared five signal representations to optimize model performance across operating conditions.
- Built an end-to-end PyTorch pipeline for preprocessing, transfer learning, and evaluation.

Open Source Contributions

- **PatchPilot** | PRs – Built a GradientBoostingClassifier-based severity ranking pipeline, selecting features and evaluating model performance, integrated into production workflows
- **localmind** | PR – Added a session cloning API endpoint for duplicating chat sessions while preserving conversation history.
- **Algo-Infinity-Verse** | PR – Built an interactive recursion learning page with visualizations, code examples, and practice exercises using HTML, CSS, and JavaScript.
- **Wagle MCP** | PR – Developed Pytest-based test cases for CLI workflows, improving reliability and validation of MCP command execution.
- View more contributions

Projects

TalentHire-AI | ***FastAPI, React, OpenAI, Qdrant, RAG, PostgreSQL*** | **GitHub** — *June 2026*

- Built an AI recruitment platform using **OpenAI GPT**, **LangChain**, and **RAG** for resume analysis, candidate matching, and interview question generation.
- Designed PostgreSQL schema and data pipeline for structured storage of parsed resumes, candidate scores, and evaluation reports.
- Built a React frontend integrating FastAPI APIs to deliver real-time candidate analysis and AI-powered hiring recommendations.

IoT Biometric Attendance System | ***ESP8266, AWS IoT Core, JavaScript*** | **GitHub** — *Aug – Dec 2024*

- Built a cloud-connected fingerprint attendance system using ESP8266 and AWS IoT Core.
- Implemented real-time attendance synchronization between IoT device and cloud backend. Designed APIs and database integration for secure attendance management.

Predictive Human Activity Modeling | ***PyTorch, GAT, BiLSTM, PyG*** | **GitHub** — *June 2025 – March 2026*

- Achieved **98% activity accuracy** and **0.93 macro F1** using a hybrid GAT + BiLSTM architecture with activity-conditioned time prediction heads across smart home sensor streams.
- Predicted time-to-next-activity with **MAE of 10–13 minutes** via quantile-based discrete time binning, stabilizing prediction over skewed inter-activity intervals.

Technical Skills

Programming Languages: Python, Java, SQL, C, HTML5, CSS3, JavaScript
Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, SDLC, Agile, Scrum, Git, GitHub
Backend & APIs: FastAPI, REST APIs
Databases: PostgreSQL, SQLite, SQL
Cloud Technologies: AWS (EC2, S3, IoT Core), Docker, Linux
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, RAG, Qdrant, NLP, LLMs
Data Processing: NumPy, Pandas, Matplotlib, Time-Series Analysis, FFT, Feature Extraction